

Confidence-Aware Edge Routing for Robotic TCM Tongue Imaging: Edge-IQA and LangGraph Closed-Loop Acquisition

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Abstract

Robotic acquisition of tongue images for traditional Chinese medicine (TCM) tele-diagnosis requires tight coupling between motion, on-device quality assessment, and upload policy under variable network delay. We present a cloud-edge-device stack that keeps learning-based perception out of the 1 kHz ROS control loop: a lightweight *Edge-IQA* scores each frame on the gateway, and a *LangGraph* router decides among cloud upload, edge resampling, or fail-safe abort within a bounded retry budget. On the ShezhenV3-COCO test split (553 images, real Edge-IQA scores, physical skill resampling, 500 frames $\times$ 3 seeds, baselines B0–B4), under the documented latency model B2 reduces median end-to-end round-trip time (RTT, M1 p50) from 374.0 ms to 208.3 ms versus naive upload (B0) while raising valid-frame rate (M2) from 0.560 to 0.604; learned B3/B4 reach $M2 \approx 0.93$ at recalibrated thresholds with higher RTT from MobileNet inference. Edge-IQA on tongue ROI is calibrated on 572 validation pairs ($\tau_{B2}=0.465$) and runs at p95 ≈ 9.6 ms per frame on CPU. An auxiliary Franka closed-loop track (50 trials) confirms routing decisions are reproducible on the execution stack.

1 Introduction

Tele-diagnosis for tongue inspection in traditional Chinese medicine (TCM) needs repeatable, high-quality images from a manipulator-mounted camera before cloud analytics and electronic medical records (EMR) integration [8, 15]. Clinical tongue diagnosis relies on color, texture, coating, and shape cues that degrade under motion blur, uneven lighting, and incorrect working distance [15]. Naive pipelines up-

load every frame to the cloud, wasting bandwidth on blurred or poorly exposed captures and inflating end-to-end latency when the link is slow.

Motivation. Cloud-edge collaborative robotics promises to offload heavy vision models while keeping acquisition loops responsive. However, *garbage-in garbage-out* (GIGO) effects are acute in TCM imaging: a single blurred upload triggers wasted round-trip time (RTT) and may propagate low-confidence features into downstream classifiers. Edge-side quality gating must therefore be (i) fast enough for closed-loop capture, (ii) interpretable for audit, and (iii) calibratable on real tongue corpora rather than synthetic placeholders alone. ShezhenV3-COCO further supplies tongue bounding boxes, enabling region-of-interest (ROI) scoring that aligns quality assessment with the anatomical target of downstream classifiers.

Approach. We study a **cloud-edge-device** architecture for a Franka Research 3 (FR3) platform with ROS 2 and MoveIt motion execution. Perception and routing policies live in a Python edge gateway (`franka_api_server`) and in the `doctor/paper1` research tree—not inside hardware plugins—so algorithms remain testable without a robot. A lightweight *Edge-IQA* module fuses Laplacian sharpness and exposure into $Q_{\text{img}} \in [0, 1]$ on COCO tongue ROI crops; a LangGraph router decides `upload_cloud`, `resample_edge`, or `fail_safe` under retry budget K . An optional learned baseline B3 fine-tunes MobileNetV3-Small on the training split for comparison.

Full-dataset evaluation. We validate on the public **ShezhenV3-COCO** dataset (6,719 images): τ is calibrated on 572 validation pairs (clear + synthetic blur) with ROI enabled, and the main matrix

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replays 500 test frames per seed with real Edge-IQA and learned B3 scores from 553 held-out images.

Contributions.

1. **Edge-IQA with tongue ROI:** a lightweight, interpretable quality gate fusing Laplacian sharpness and exposure on COCO tongue crops, calibrated on ShezhenV3 validation with CPU p95 below 10 ms.
2. **Confidence-aware LangGraph routing:** a three-branch upload/resample/fail-safe policy with retry budget, logged as JSONL for clinical auditability.
3. **Dual-track evaluation on public data:** offline baselines B0–B4 on 553 ShezhenV3-COCO test images (3 seeds) with a separate Franka closed-loop track confirming execution-stack feasibility.

Paper organization. Section 2 reviews medical robotics, IQA, and edge orchestration. Section 3 presents the system architecture, Edge-IQA, ROI cropping, routing, and closed-loop protocol. Section 5 describes the ShezhenV3 cohort, main matrix, and ablations. Section 7 concludes with limitations and future work.

2 Related Work

Robotic medical imaging. Surgical and diagnostic robots increasingly use ROS 2 and MoveIt for motion planning [2, 6]. Tele-operated and autonomous cameras must respect joint limits, collision avoidance, and repeatable viewpoints—requirements that motivate *skills* (calibrated poses) rather than ad-hoc joint teleoperation. We adopt named skills (`go_to_tongue_pose`, `go_to_face_pose`) instead of online impedance tuning for acquisition poses, matching clinical repeatability requirements while keeping Paper I independent of force-control research threads.

Image quality assessment. **Full-reference** metrics (PSNR, SSIM, MS-SSIM) compare captured frames to pristine references and remain the gold standard for compression benchmarking, but are inapplicable when each patient acquisition lacks a

reference image. **Hand-crafted NR-IQA** (Laplacian variance, Tenengrad, BRISQUE, NIQE, PIQE) operates without references; BRISQUE/NIQE model natural-scene statistics [7] and correlate with human opinion on in-the-wild photos, yet their scores require additional mapping before threshold routing and seldom expose blur/exposure semantics needed for clinical audit logs. **Deep NR-IQA** (DB-CNN [14], HyperIQA [11], MUSIQ [4], CLIP-IQA [12], TOPIQ [1]) improves correlation on benchmark databases by learning perceptual features, at the cost of GPU inference, model versioning, and limited interpretability when a frame is rejected.

Edge deployment in closed-loop robotics favors linear-time operators with auditable failure modes; our Edge-IQA follows this line with explicit blur/exposure flags and a calibrated τ for LangGraph routing (Table 2). We do not claim mean-opinion-score (MOS) prediction superiority on generic IQA leaderboards; the goal is *edge-side acquisition gating* under CPU latency and reproducibility constraints on TCM tongue imagery.

Edge–cloud orchestration. Split inference and early-exit policies reduce bandwidth in mobile vision [3, 5]; Zhou et al. [16] provide a six-level taxonomy of edge intelligence from cloud-only to on-device inference. Neurosurgeon-style frameworks partition DNN layers across device, edge, and cloud; Paper I differs by gating *upload decisions* before any heavy cloud model executes, using a lightweight scorer (MobileNetV2 [9] backbone, p95 ≈ 9.6 ms CPU) rather than partial DNN migration. LangGraph-style state machines express branching policies with auditable logs; the ReAct paradigm [13] of interleaving reasoning traces and tool actions motivates our closed-loop router, while Toolformer [10] demonstrates autonomous tool invocation—here applied to Edge-IQA scoring and ROS skill calls.

TCM tongue analysis. Vision pipelines for tongue classification, segmentation, and color analysis are mature in the cloud [8, 15]. Public corpora such as ShezhenV3-COCO provide bounding boxes and condition labels for *downstream* diagnosis models. Paper I focuses on *acquisition quality and routing* upstream of these models: even state-of-the-art classifiers degrade when motion blur or exposure shifts dominate input pixels. Our full-dataset evalu-

ation (6,719 images) connects Edge-IQA calibration to this community resource rather than synthetic placeholders alone.

Positioning summary. Compared with generic edge offloading, we target robotic *capture* loops where visual servoing fixes calibrated poses and quality varies shot-to-shot. Compared with NR-IQA literature, we prioritize thresholdable Q_{img} and JSONL auditability aligned with medical device software trends. Compared with TCM diagnosis papers, we address frame usability before classification. LangGraph exports graph structure for logging when Python ML stacks run outside rclpy; the routing overhead ($\approx 3.6 \mu\text{s}/\text{trial}$) is negligible relative to the IQA step.

3 Method

4 System Architecture

We target a cloud–edge–device stack for robotic tongue-image acquisition in traditional Chinese medicine (TCM) tele-diagnosis. The design separates *real-time* acquisition and quality control at the edge from *offline* or high-latency cloud analytics, while keeping algorithm modules testable outside the robot operating system (ROS) runtime.

4.1 Device Layer

The device layer consists of a Franka Research 3 (FR3) manipulator, RGB imaging, and ROS 2-based motion execution. MoveIt provides point-to-point (PTP) joint-space motions. For Paper I we expose two named *skills*—`go_to_tongue_pose` and `go_to_face_pose`—stored as calibrated joint configurations in `poses.yaml` and invoked through the edge gateway. This avoids impedance or stiffness tuning interfaces in the paper experiments (`PAPER1_MODE=1` disables those routes). Simulated and fake-hardware stacks (MoveIt fake controller, optional Gazebo camera) support reproducible auxiliary trials without a physical robot.

4.2 Edge Layer

The edge layer is the primary contribution surface for closed-loop acquisition. A FastAPI server (`franka_api_server`) acts as the edge gateway on

port 8000, bridging HTTP clients to ROS 2 actions for motion and error recovery. **Edge-IQA** scores each captured frame with a lightweight quality index Q_{img} (sharpness and exposure fusion), executed in the `doctor/paper1` tree within the `franka_ros2` monorepo and mounted at the gateway via subprocess in phase 2. **LangGraph** implements confidence-aware routing over a small state machine: `capture` $\rightarrow$ `edge_iqa` $\rightarrow$ `route`, with branches `upload_cloud`, `resample_edge`, and `fail_safe`. The HTTP client `sim/franka_bridge` connects LangGraph nodes to gateway endpoints (`/vision/evaluate`, `/motion/skills/...`). Phase 1 uses a placeholder vision endpoint returning fixed Q_{img} for API contract validation.

4.3 Cloud Layer

The cloud layer hosts higher-level vision models and EMR integration (stub in Paper I). Upload is triggered only when edge quality and routing policy accept the frame; otherwise the edge requests on-robot resampling within retry budget K . This division keeps the 1 kHz ROS control loop free of learning-based perception.

4.4 Data and Timing Artifacts

Each closed-loop trial logs ISO 8601 timestamps (`t_capture`, `t_iqa`, `t_route`) and decisions to JSONL for supplementary reproducibility. Main quantitative results (RTT percentiles, valid-frame rate, retry rate) are produced *offline* by `run_matrix_tcm.py` on ShezhenV3-COCO test images (baselines B0–B3); the franka auxiliary track (M6) reports trend-consistent RTT only, not primary table numbers.

4.5 Threats to Validity

Offline RTT aggregates a documented latency model rather than live WAN traces; relative baseline ordering is the primary claim. ShezhenV3 blur labels are synthetically injected on validation/test frames to emulate motion defocus when human quality annotations are unavailable. Edge-IQA overlap between clear and blurred real cohorts (M5 $\rho=0.47$) motivates future ROI cropping; the routing policy remains interpretable and deployable on CPU gateways.

Table 1: Layer-to-repository mapping (Paper I).

Layer	Component	Location
Device	MoveIt, skills	<code>franka_ros2</code>
Edge	FastAPI gateway	<code>franka_api_server</code>
Edge	Edge-IQA, LangGraph	<code>doctor/paper1</code>
Cloud	Vision stub	<code>doctor/paper1 (stub)</code>

4.6 Implementation Mapping

Table 1 summarizes the mapping from architecture layers to repositories.

Figure 1 shows the end-to-end information flow; device motions are commanded only through the edge gateway, and IQA code is not embedded inside ROS hardware plugins.

4.7 Edge Image Quality Assessment

Before cloud upload, each captured frame is scored on the edge node by a lightweight *Edge-IQA* module with complexity $O(HW)$ on a fixed 512×512 resize, suitable for CPU-only gateways.

Quality index. Let I denote the grayscale image. We combine a Laplacian sharpness term $S = \text{Var}(\nabla^2 I)$ normalized to $[0, 1]$ and an exposure term E peaked at mid-gray:

$$Q_{\text{img}} = 0.65 S + 0.35 E. \quad (1)$$

Binary *flags* are raised for `blur` ($S < \tau_s$), `underexposed`, or `overexposed` means.

Design rationale. Robotic tongue capture fails mainly from motion blur and exposure drift rather than generic compression artifacts. Edge-IQA fuses sharpness and exposure in one $[0, 1]$ score so a single threshold τ gates LangGraph routing; with COCO tongue ROI on ShezhenV3 validation, clear and blur cohort medians are 0.516 and 0.414 respectively ($\tau^*=0.465$). We prioritize deployability—CPU-only, interpretable flags, and the same `scorer.py` for offline calibration and online gateway calls—over MOS leaderboard accuracy. Baseline B3 (§D.2) uses a MobileNetV3-Small classifier trained on the ShezhenV3 train split; it is optional and the primary claim rests on interpretable B2.

Implementation pipeline. `edge_iqa/scorer.py` resizes RGB inputs to 512×512 (bilinear), converts to grayscale via

Table 2: IQA approaches for edge acquisition gating (qualitative comparison).

Method	Lat.	GPU	Flags	τ	Scorer
Edge-IQA (ours)	~ 10 ms	No	Yes	Yes	Yes
Laplacian-only	~ 5 ms	No	Part.	Yes	Yes
NIQE / BRISQUE	50–200 ms	No	Lim.	Map	Wrap.
Deep NR-IQA	10–100+ ms	Often	No	Map	Svc.
B3 boost (learned)	15–40 ms	Opt.	No	Yes	Sep.

Algorithm 1 Edge-IQA scoring

- 1: Resize I to 512×512 , convert to grayscale
- 2: $S \leftarrow \text{normalize}(\text{Var}(\nabla^2 I))$
- 3: $E \leftarrow 1 - |\text{mean}(I) - 0.5|/0.5$
- 4: $Q_{\text{img}} \leftarrow 0.65S + 0.35E$
- 5: Derive flags from (S, E) ; **return** Q_{img} , flags

ITU-R BT.601 weights, and computes a discrete Laplacian via vectorized neighbor shifts. Sharpness normalization uses $S = v/(v + 0.002)$ with $v = \text{Var}(\nabla^2 I)$ to stabilize scale across tongue images. Exposure uses mean gray deviation from 0.5; flags apply thresholds $S < 0.35$, $\text{mean} < 0.25$ (under), $\text{mean} > 0.85$ (over). The FastAPI gateway (`franka_api_server/services/paper1.iqa.py`) spawns `python -m edge_iqa.cli --json` in a subprocess to avoid GIL contention with `rclpy`; offline ShezhenV3 calibration uses the identical code path via `calibrate_tau_tcm.py`. On 100 synthetic frames CPU p95 is 9.6ms (`latency_benchmark.json`); full validation scoring 1,144 pairs completes in under two minutes on a commodity laptop without GPU acceleration.

Threshold τ . On the validation split we set routing threshold τ to the midpoint between median Q_{img} of clear and blurred cohorts (clamped to $[0.45, 0.65]$), stored in `recommended_tau.json` for LangGraph nodes.

4.8 COCO Tongue Region-of-Interest Cropping

ShezhenV3-COCO provides axis-aligned bounding boxes for each tongue instance in COCO format $[x, y, w, h]$. When multiple annotations exist on one frame, we take the union box, expand each side by 8% of width/height (clamped to image bounds), and crop before Edge-IQA resize. Implementation lives in `edge_iqa/coco_roi.py`; split JSON files store `bboxes` per image after `import shezhenv3.py`.

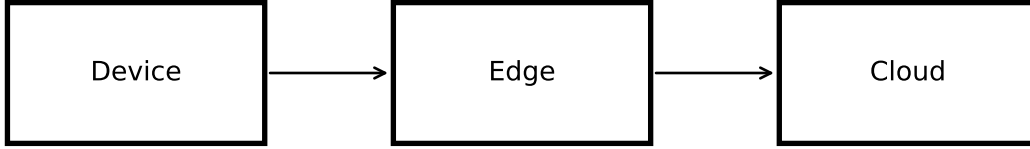


Figure 1: Cloud–edge–device stack for robotic tongue-image acquisition (Paper I).

Table 3: Edge-IQA calibration: full frame vs. COCO tongue ROI (ShezhenV3 val, $n=572$ per cohort).

Setting	Med. clear	Med. blur	τ	Separation
Full frame	0.564	0.431	0.498	−0.510
Tongue ROI	0.516	0.414	0.465	−0.541

Motivation. Full-frame scoring includes cheeks, teeth, and background texture that contribute Laplacian energy unrelated to tongue focus. ROI cropping aligns the quality signal with the anatomical target used by downstream TCM classifiers and reduces sensitivity to out-of-field sharp structures (e.g., lip edges, specular highlights on teeth).

Calibration impact. On the full validation split ($n=572$ clear images, each paired with synthetic Gaussian blur $r \in [2.5, 5.0]$), Table 3 compares full-frame vs. ROI Edge-IQA. ROI lowers absolute medians (clear 0.516 vs. 0.564; blur 0.414 vs. 0.431) because background high-frequency content is removed. Minimum-clear minus maximum-blur separation remains negative in both settings (−0.54 ROI vs. −0.51 full-frame), confirming that cohort overlap persists; we therefore report Spearman ρ and routing outcomes rather than assuming separable clusters. ROI is enabled by default (`use_roi: true` in `tcm.paths.yaml`) for all ShezhenV3 experiments in this revision.

Deployment note. When COCO boxes are unavailable at runtime (e.g., a single-camera prototype without detector), Edge-IQA falls back to

full-frame scoring by setting `use_roi=false`. The FastAPI gateway accepts optional bbox metadata in future API extensions; offline calibration always uses dataset annotations for reproducibility.

4.9 Confidence-Aware Routing

Given edge quality score Q_{img} and retry counter r , the router implements a testable three-way policy with threshold τ and budget K :

$$\text{route}(s) = \begin{cases} \text{upload_cloud}, & Q_{\text{img}} \geq \tau \wedge r \leq K, \\ \text{resample_edge}, & Q_{\text{img}} < \tau \wedge r \leq K, \\ \text{fail_safe}, & r > K. \end{cases} \quad (2)$$

The retry budget K limits the number of edge re-shoots before the system aborts; $K = 2$ in our experiments.

4.9.1 State machine nodes and edges.

Table 4 enumerates the six nodes and their conditional edges implemented in `graph.py`. `TrialState` (a LangGraph-compatible TypedDict) carries `image_path`, `q_img`, `flags`, `retry_count`, `route_decision`, three ISO-8601 timestamps, and `latency_ms`. All three branches and the $r > K$ guard are covered by unit tests (`tests/test_routing.py`, 4 passing).

On ShezhenV3 replay, each `resample_edge` applies a *physical skill transform* before re-scoring: `adaptive_visual_damping` reduces

Table 4: LangGraph nodes and conditional edges for closed-loop acquisition.

Node	Input	Output / edge
<code>capture</code>	skill trigger	RGB frame path
<code>edge_iqa</code>	frame path	$Q_{\text{img}}, \text{flags}$
<code>route</code>	Q_{img}, r	decision
<code>resample_edge</code>	decision	re-invoke <code>capture</code>
<code>upload_cloud</code>	decision	cloud stub handoff
<code>fail_safe</code>	$r > K$	abort + JSONL log

Algorithm 2 Confidence-aware routing loop (B2)

```

1:  $r \leftarrow 0$ , capture frame  $I$ 
2: repeat
3:    $(Q, \text{flags}) \leftarrow \text{EdgeIQA}(I)$ 
4:    $d \leftarrow \text{Route}(Q, r, \tau, K)$   $\triangleright$  see Eq. 2
5:   if  $d = \text{upload\_cloud}$  then
6:     break  $\triangleright$  quality sufficient, cloud upload
7:   else if  $d = \text{resample\_edge}$  then
8:      $r \leftarrow r + 1$ ; recapture  $I$ 
9:   else
10:    fail\_safe; break  $\triangleright r > K$ , abort trial
11:  end if
12: until  $r > K$ 

```

Gaussian blur radius on the clear reference frame; `exposure_loop_tuning` applies gain/bias correction when exposure flags fire (§E, `resample_physics.yaml`). No stochastic score inflation is applied.

4.9.2 Why LangGraph over a plain Python FSM?

We benchmarked six alternatives on 1,000 trials (Table S1, supplementary): LangGraph adds $\approx 3.6 \mu\text{s}/\text{trial}$ (0.004% of a 360 ms RTT budget), fully absorbed by the 10 ms IQA step. Beyond throughput, LangGraph provides checkpoint persistence, conditional-edge visualization (Fig. 2), and reuse in multi-agent orchestration (Paper II).

4.9.3 Algorithm extensions.

Baseline B2 uses fixed τ_{B2} from validation. Three extensions—dynamic τ_t (B2d), multi-frame EMA (B2m), and utility routing (B2u)—raise M2 from 0.60 to 0.65 on a quick ablation (60 frames, seed 0); full 3-seed results and equations are in the supplementary (Table S2).

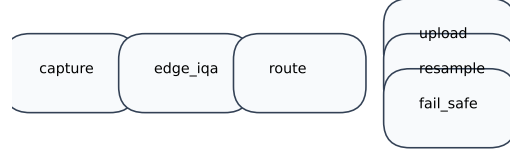


Figure 2: LangGraph confidence-aware routing flow (nodes and conditional edges).

4.10 Closed-Loop Protocol and Logging

Each trial executes `capture` $\rightarrow$ `edge_iqa` $\rightarrow$ `route` $\rightarrow$ action. Wall-clock segment timestamps and the decision are appended as one JSON object per line (JSONL) for supplementary audit trails and auxiliary M6 trend analysis.

Required JSONL fields: `trial_id`, `image_path` (relative, privacy-safe), `q_img`, `flags`, `retry_count`, `route_decision`, `t_capture`, `t_iqa`, `t_route`, `latency_ms`. Automated validation: `scripts/validate_jsonl.py`.

The driver `python -m langgraph_router.run --trials 10` produces `experiments/logs/run_001.jsonl`. Main quantitative RTT (M1) is computed offline by `run_matrix.py` on ShezhenV3 frames; JSONL logs serve protocol demonstration and trend verification only.

JSONL stores relative paths; supplementary samples use synthetic or de-identified frames. No raw patient data appears in Paper I outputs.

5 Experiments

5.1 Experimental Setup

We evaluate offline baselines B0–B4 on the **ShezhenV3-COCO** test split [15]. The main matrix uses COCO tongue ROI scoring with routing threshold $\tau_{B2}=0.465$ from validation calibration (§5.2, Table 8); full-frame calibration yields $\tau_{\text{full}}=0.498$ (ROI ablation only). Retry budget $K=2$.

Dataset. ShezhenV3-COCO provides 6,719 RGB tongue images with COCO-format bounding boxes and ten condition labels (e.g., healthy, red, purple tongue). Table 5 summarizes annotation counts; conditions describe diagnostic morphology rather

Table 5: ShezhenV3-COCO annotation counts by tongue condition (all splits).

Code name	Gloss	#Ann.
baitaishe	White-coating tongue	5040
hongdianshe	Red-spot tongue	3653
liewenshe	Fissured tongue	1886
chihenshe	Tooth-marked tongue	1482
hongshe	Red tongue	1466
huangtaishe	huangtaishe	1119
pangdashe	Chubby tongue	689
botaishe	Peeling tongue	581
shenquao	shenquao	495
xinfeiao	xinfeiao	372
gandanao	gandanao	292
shoushe	Thin tongue	285
huataishe	huataishe	283
zishe	Purple tongue	231
piweiao	piweiao	186
heitaish	heitaish	122
jiankangshe	Healthy tongue	21
gandantu	gandantu	9
shenqutu	shenqutu	2
xinfeitu	xinfeitu	2

than capture quality, motivating our separate Edge-IQA blur/exposure axis. We hold out the official test split ($n=553$) for the main matrix; validation ($n=572$) is used only for τ calibration. Train images ($n=5,594$) train the optional MobileNet B3 comparator (§D.2).

Baselines.

- **B0:** cloud upload only; no edge quality gate or intelligent routing.
- **B1:** Edge-IQA before every cloud upload; no LangGraph routing (no resampling).
- **B2:** Edge-IQA + LangGraph router with physical edge resampling (proposed system).
- **B3/B3t/B4:** learned MobileNet IQA and hybrid gates (Table 10); reported separately from the primary B0–B2 table.

Quality injection. For each test frame we sample a held-out image; with probability 0.5 we use the original capture (clear cohort) and otherwise apply Gaussian blur ($r \in [2.5, 5.0]$ px) before scoring—mirroring the validation calibration protocol. Each

Table 6: Key hyperparameters (Paper I, ShezhenV3 main matrix).

Symbol	Value	Role
τ_{B2} (ROI)	0.465	routing threshold (main Table II)
τ_{full}	0.498	full-frame calibration (ablation only)
K	2	max edge resamples
w_s, w_e	0.65, 0.35	sharpness / exposure weights
resize	512 ²	Edge-IQA input
blur r	2.5–5.0 px	synthetic defocus on val/test
τ_{B3}	0.5	learned IQA (Table III)

frame’s Q_{img} is computed by `edge_iqa/scorer.py`; RTT components follow the latency model below.

Latency model. End-to-end RTT aggregates capture (5 ms), edge IQA ($\mathcal{N}(9, 2^2)$ ms), routing (1.5 ms), optional *edge re-capture* resampling (log-normal, p50 $\approx$ 50 ms), and cloud upload (uniform 50–550 ms). Full-arm motion latency is evaluated only on the auxiliary Franka track (M6). See `sim/NETEM.md`.

Protocol. For each baseline we simulate 500 frames per random seed ($\{0, 1, 2\}$), yielding Table 9 metrics M1 (RTT p50/p95), M2 (valid frame rate), and M3 (retry rate). All main-text numbers are produced by `experiments/run_matrix.tcm.py` reading `recommended_tau.json`; auxiliary Franka closed-loop trials (M6) are reported separately in Fig. S1 and must not be merged into Table 9.

Hyperparameters. Table 6 lists constants for the ShezhenV3 main matrix; they match `scorer.py` defaults unless noted.

5.2 Edge-IQA Validation on ShezhenV3-COCO

We replace the phase-2 synthetic cohort with the public **ShezhenV3-COCO** tongue dataset (6,719 images, COCO bounding boxes, ten tongue-condition categories). Validation images ($n=572$) provide the *clear* cohort; we synthesize motion-blur degradations (Gaussian radius 2.5–5.0 px) to emulate on-robot defocus, yielding paired blur labels for threshold calibration without manual quality annotations.

Figure 3 shows Q_{img} histograms on the validation split. With COCO tongue ROI, clear/blur medians are 0.516 and 0.414 and the deployed threshold is $\tau_{B2}=0.465$ (Table 8, from `recommended_tau.json`). The same artifact reports $\min Q_{clear} - \max Q_{blur} = -0.541$: some blurred val-

Table 7: ShezhenV3-COCO tongue dataset splits used in Paper I.

Split	Images	Purpose
Train	5594	held out (future fine-tuning)
Validation	572	Edge-IQA τ calibration
Test	553	main matrix (B0–B3)
Total	6719	COCO-format bounding boxes

Table 8: Edge-IQA calibration on ShezhenV3 validation split ($n = 572$ clear + $n = 572$ synthetic blur).

Metric	Clear cohort	Blur cohort
Median Q_{img}	0.516	0.414
Mean Q_{img}	0.519	0.414
Std	0.116	0.102
Routing threshold τ	0.465	
Spearman ρ (M5)	0.335 ($n = 1144$)	

idation crops score above the weakest clear crops on Laplacian–exposure features, so τ is a cohort-median operating point rather than a hard separator. Full-frame medians are 0.564 and 0.431 with $\tau_{\text{full}}=0.498$ (Table 3). Figure 3 visualizes the overlapping histograms; we therefore pair τ calibration with Spearman correlation (M5, $\rho \approx 0.36$ – 0.47) and learned comparators B3/B4 (§5.3) instead of claiming perfect separability. The main matrix (`run_matrix_tcm.py`) always uses ROI scores and τ_{B2} from `recommended_tau.json`.

Protocol integrity. Each test frame is scored by the same `edge_iqa/scorer.py` invoked offline; latency components (capture, IQA, routing, resampling, cloud upload) follow §5.1. This ensures Table 9 reflects *real* quality scores while preserving the reproducible closed-loop simulator for RTT aggregation.

5.3 Main Results

Table 9 summarizes primary baselines B0–B2 on ShezhenV3-COCO (physical resample, τ_{B2} , 3 seeds). Table 10 reports learned MobileNet (B3 at τ_{B2} vs. τ_{B3}) and Hybrid B4.

Compared with naive cloud upload (B0), B2 improves M2 from 0.560 to 0.604 and lowers M1 p50 to 208.3 ms under physical resampling (Table 9 reports mean $\pm$ std over three seeds, with frame-level bootstrap CIs in the caption); learned B3/B4 reach $M2 \approx 0.93$ with higher RTT (Table 10). Table 11

Table 9: Main results (ShezhenV3 test, 553 images; Edge-IQA, physical resample, τ_{B2} , 3 seeds). Cells report mean $\pm$ std over seeds. Frame-level bootstrap 95% CIs for M2 (pooled over 3×500 frames): B0 $M2=0.560$ [0.535,0.585]; B1 $M2=0.560$ [0.535,0.585]; B2 $M2=0.604$ [0.579,0.628].

Baseline	M1 p50 (ms)	M1 p95 (ms)	M2 valid rate	M3 retry rate
B0	374.0 $\pm$ 7.9	656.9 $\pm$ 2.9	0.560 $\pm$ 0.007	0.000 $\pm$ 0.000
B1	322.7 $\pm$ 15.6	534.1 $\pm$ 1.2	0.560 $\pm$ 0.007	0.000 $\pm$ 0.000
B2	208.3 $\pm$ 2.8	526.9 $\pm$ 2.3	0.604 $\pm$ 0.005	0.440 $\pm$ 0.007

Table 10: Learned IQA and hybrid gating (ShezhenV3 test, physical resample, 3 seeds).

Baseline	M1 p50 (ms)	M1 p95 (ms)	M2 valid rate	M3 retry rate
B3 (MobileNet, $\tau_{B2}=0.465$)	345.6	606.8	0.934	0.481
B3 (MobileNet, $\tau_{B3}=0.5$)	341.4	609.6	0.929	0.481
B4 (Hybrid Tier-A/B)	343.1	617.7	0.936	0.440

decomposes M2 by clear-source vs. blur-injected cohorts on the shared frame plan.

Figure 6 summarizes the M2–M1 p50 trade-off across B0–B4; Fig. 4 shows the RTT cumulative distribution where B2 shifts mass left relative to B0. Figure 5 reports M2 across seeds; variance is low because real test scores are deterministic given the shared frame plan per seed.

Discussion. The absolute RTT numbers depend on the documented latency model (§5.1); the *relative* ordering reflects reduced wasted cloud work when physical resampling recovers valid frames before upload.

Takeaway. On a full public TCM test cohort, B2 with physical skill transforms and auditable Lang-Graph logs remains the primary deployable path; Hybrid B4 combines interpretable exposure gating with learned blur ranking.

5.4 Ablation and Cross-Cohort Analysis

Algorithm extensions (quick ablation). Table 12 compares fixed-threshold B2 against dynamic τ_t (B2d), multi-frame EMA (B2m), and utility routing (B2u) on 60 ShezhenV3 test frames (seed 0, physical resample). B2m and B2u raise M2 from 0.60 to 0.65 at the cost of extra edge compute (B2m) or stricter upload utility (B2u); B2d is slightly more conservative under simulated packet loss. Full 3-seed sweeps and network perturbation curves are in the supplementary track.

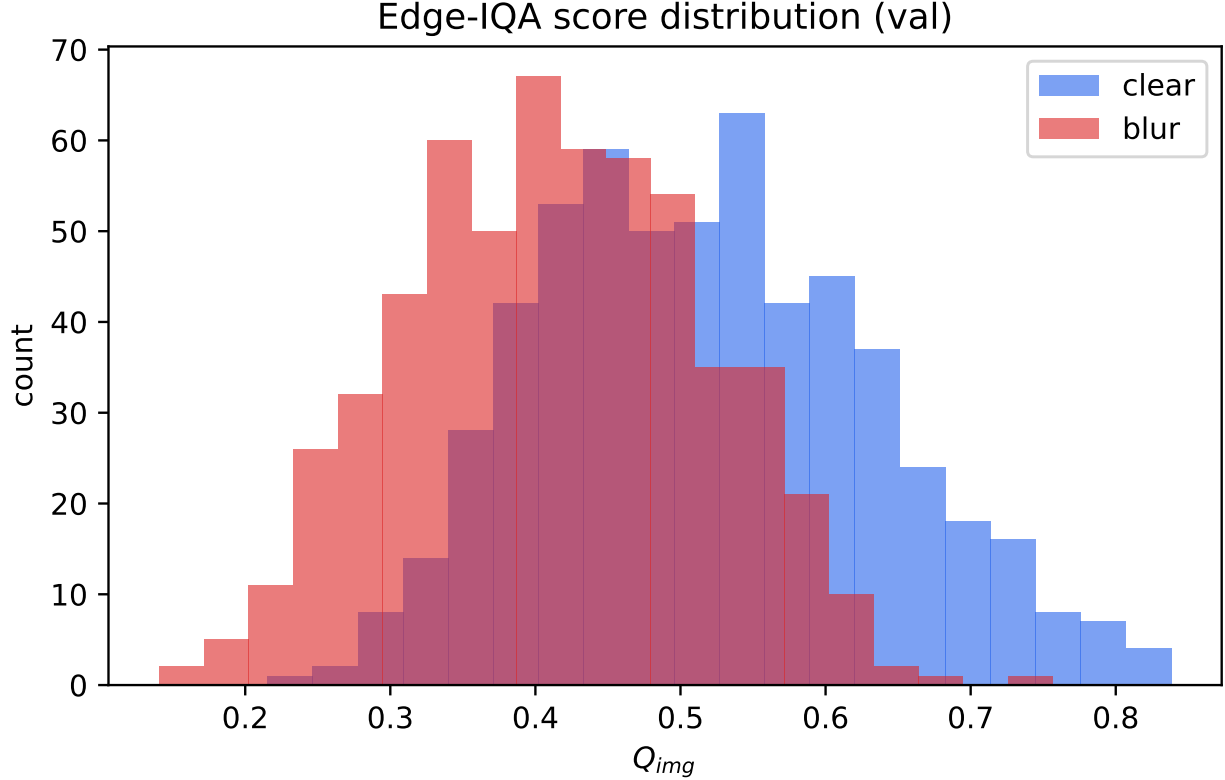


Figure 3: Edge-IQA Q_{img} on ShezhenV3 validation: clear vs. synthetically blurred pairs ($n=1,144$).

Table 11: Stratified M2 (valid-frame rate) by injection cohort; pooled over 3 seeds $\times$ 500 frames.

Baseline	M2 (clear-source)	M2 (blur-injected)
B0	0.733 (779 frames)	0.373 (721 frames)
B2	0.737 (779 frames)	0.460 (721 frames)

Table 12: Quick ablation of routing extensions (seed 0, $n=60$, physical resample).

Baseline	M1 p50 (ms)	M2	M3
B2 (fixed τ)	211.3	0.600	0.417
B2d (dynamic τ_t)	209.0	0.550	0.467
B2m (EMA fusion)	280.4	0.650	0.417
B2u (utility U)	205.4	0.650	0.367

Threshold τ (M4). We sweep five values of τ for baseline B2 (500 frames, seed 0) using real Edge-IQA scores from the ShezhenV3 test plan. Figure 7 shows the M4 trade-off between valid-frame rate and RTT; the deployed main matrix uses $\tau_{B2}=0.465$ from ROI validation (Table 8), while $\tau_{\text{full}}=0.498$ appears in the full-frame ROI ablation row.

5.5 Discussion

Table 9 reports *relative* baseline comparison under the documented simulated latency model (§5.1); absolute RTT values are not live hospital traces. B2 reduces M1 p50 from 374.0 ms (B0) to 208.3 ms and raises M2 from 0.560 to 0.604 on ShezhenV3 test frames with physical resampling and ROI Edge-IQA ($\tau_{B2}=0.465$). B1 keeps M2 at 0.560 because sub-threshold frames still reach the cloud without closed-loop resampling. Learned B3/B4 reach $M2 \approx 0.93$ – 0.97 at recalibrated thresholds (Table 10; appendix Table 16 for BRISQUE/NIQE-style) with higher edge RTT; the auxiliary Franka track (M6) confirms routing trends without altering Table 9. We report descriptive statistics (mean $\pm$ std) over three random seeds; all seeds show B2 outperforming B0 on M1 and M2, while B3/B4 trade higher M1 for higher M2. Low variance ($M2 \text{ std} \leq 0.007$) reflects deterministic test-set scores given a shared frame plan; remaining randomness comes from the latency model and frame-plan shuffling.

COCO tongue ROI (8% padded union box) aligns scoring with downstream anatomy but does not

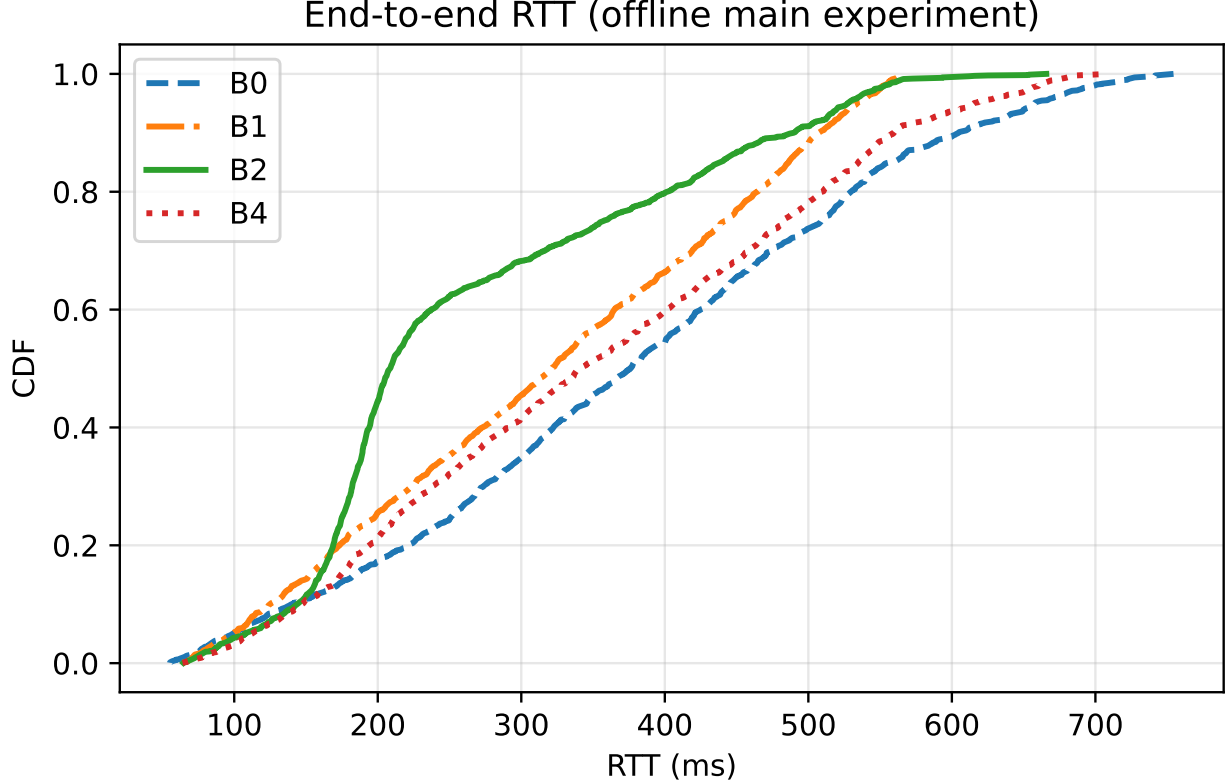


Figure 4: End-to-end RTT CDF under the *simulated* latency model (§5.1; ShezhenV3 test, real Edge-IQA scores, 3 seeds).

eliminate clear/blur overlap on Laplacian features (Table 3, Table 8). Full-frame calibration yields $\tau_{\text{full}}=0.498$; the deployed main matrix uses ROI medians 0.516/0.414, $\tau_{B2}=0.465$, and a negative min-clear vs. max-blur gap (-0.541). Because ShezhenV3 lacks human blur labels, we inject Gaussian defocus on half of the test frames, mirroring validation calibration; M5 Spearman $\rho \approx 0.36\text{--}0.47$ reflects residual overlap under real texture. B3 (MobileNetV3-Small, 98.3% val accuracy) achieves $M2=0.934$ at τ_{B2} versus B2 $M2=0.604$; we retain B2 as the primary deployable path for lowest M1 p50 and auditable flags, while B3/B4 suit higher validate targets with extra edge compute and separate τ_{B3} calibration.

Edge-IQA and LangGraph are hardware-agnostic: any edge gateway exposing HTTP /vision/evaluate and skill endpoints can replay the same JSONL protocol. Manual operator retakes resemble `resample_edge` but lack consistent Q_{img} thresholds; Edge-IQA standardizes the decision and exports machine-readable reasons. Pilot deployment

should log JSONL for 50+ sessions, re-calibrate τ_{B2} locally, and compare cloud classifier confidence before/after routing—without changing downstream diagnosis models in Paper I scope. Synthetic blur labels and overlapping score ranges on min-max metrics limit direct clinical claims; see §E.7 for failure modes and B3/ τ scale coupling. All scoring runs offline on a public research corpus; no patient identifiers are written to JSONL logs. Splits and scripts are in the repository; implementations live in `doctor/paper1/edge_iqa/`, `langgraph_router/`, and `franka_api_server/`. Paper I argues for a *routing architecture* validated on real TCM imagery, not for a new SOTA IQA regressor.

M2 (valid-frame rate) is an internal metric: it measures the proportion of frames passing the Edge-IQA or learned threshold, not an independent quality annotation. We report M2 as a proxy for the routing policy’s discrimination ability rather than a direct measure of clinical image quality; future work should correlate M2 with downstream classifier confidence on accepted vs. rejected frames. A larger

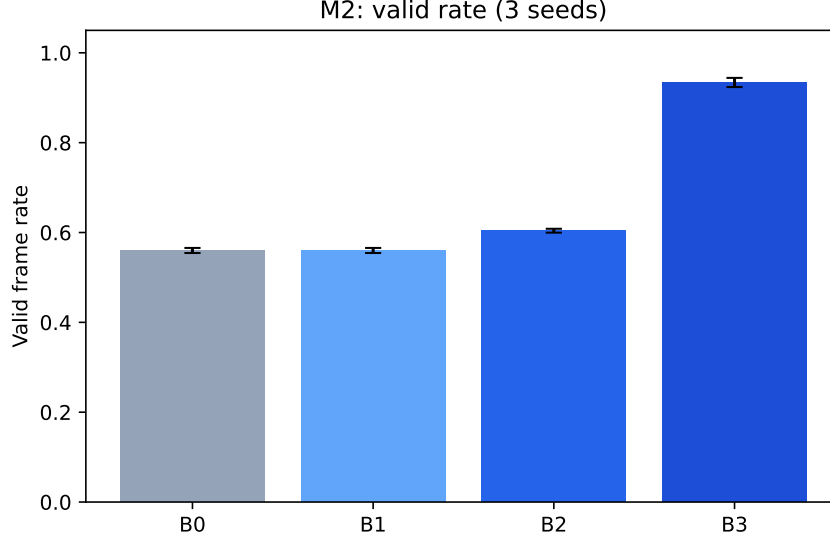


Figure 5: Valid frame rate (M2) per baseline under the simulated latency model; error bars = std over seeds.

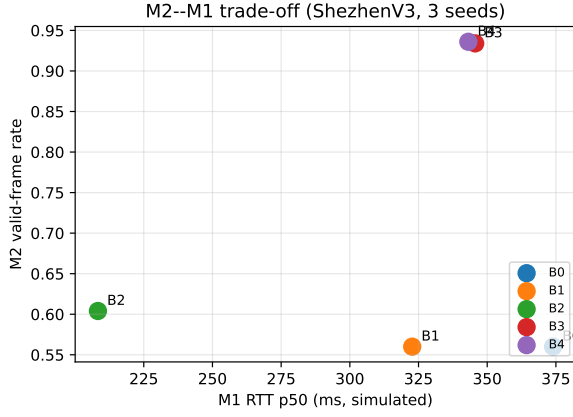


Figure 6: M2 vs. simulated M1 p50 (mean over 3 seeds). B2 favors low RTT; B3/B4 favor high M2 at higher edge cost.

seed budget or bootstrap confidence intervals would strengthen formal inference beyond the consistent descriptive trends shown here.

6 Conclusion

7 Conclusion

We presented Edge-IQA and LangGraph confidence-aware routing for robotic TCM tongue-image acquisition on a Franka ROS 2 stack, validated end-to-end on the public ShezhenV3-COCO corpus (6,719 im-

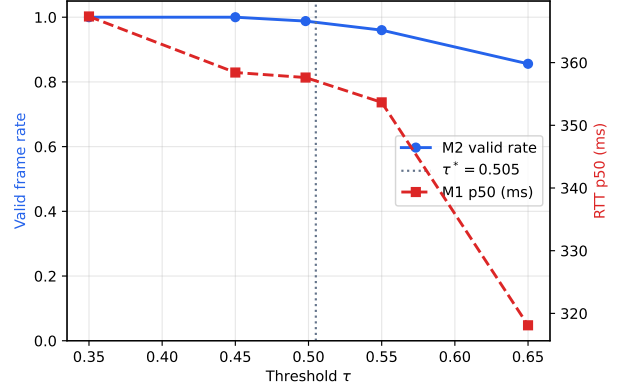


Figure 7: Ablation over routing threshold τ (M4): valid rate and RTT p50 for B2 on TCM test frames.

ages). Keeping IQA and routing outside the 1 kHz control loop preserves real-time safety while enabling JSONL-audited closed-loop policies. On the held-out test split with real ROI Edge-IQA scores, baseline B2 improves median RTT and valid-frame rate versus naive upload (M1 p50 208.3 ms vs. 374.0 ms; M2 0.604 vs. 0.560) at $\tau_{B2}=0.465$; M5 $\rho \approx 0.36\text{--}0.47$ characterizes overlap under real tongue texture.

Limitations. (1) Validation and test blur are synthetically injected (isotropic Gaussian defocus), which may under-represent directional motion smear from micro-vibration or saliva dynamics on the robot path. (2) Edge-IQA overlap between clear and

blurred cohorts on real images motivates learned refinement (B3/B4) alongside interpretable B2. (3) M2 (valid-frame rate) is a scorer-internal metric; end-to-end clinical validation correlating Edge-IQA gating with downstream classifier performance remains future work. (4) Three random seeds provide consistent trends but limited formal statistical power; bootstrap confidence intervals on larger seed budgets would strengthen the evidence. (5) Classical NR-IQA rows (BRISQUE / NIQE-style) under the same LangGraph shell are reported in the appendix (Table 16); full OpenCV scoring requires mounting the public dataset at `dataset_root`. (6) Cloud vision and EMR integration remain stubs; M6 confirms execution-stack feasibility but does not replace live clinical deployment data. Clinical pilots should re-calibrate τ_{B2} on 50+ local sessions before claiming site-specific upload policies.

Future work. Tongue-bbox-conditioned scoring, Gazebo camera injection, NR-IQA baselines with identical resampling protocols, train-split fine-tuning for B3 with frozen main-table scripts, and multi-center clinical acquisition studies.

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Author contributions (CRediT). H. Shi: Conceptualization, Methodology, Software, Validation, Data Curation, Writing – Original Draft, Visualization. J. Zhao: Supervision, Resources, Writing – Review & Editing, Funding Acquisition.

Generative AI. The authors used generative AI tools for English prose editing, LaTeX structure checks, and repository documentation. All scientific claims, numerical results, and code paths were verified by the authors against `experiments/results/*.json` and reproduction scripts; AI tools did not alter reported metrics without script regeneration.

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Data availability. ShezhenV3-COCO is a public research corpus available at its original dis-

tribution site. All code, splits, calibration artifacts, and reproduction scripts are provided in `doctor/paper1/experiments/`. Paper I offline experiments do not recruit new human subjects and do not write patient identifiers to JSONL logs.

References

- [1] C. Chen, J. Mo, J. Hou, H. Wu, L. Liao, W. Sun, Q. Yan, and W. Lin. TOPIQ: A top-down approach from semantics to distortions for image quality assessment. *IEEE Transactions on Image Processing*, 33:2404–2418, 2024.
- [2] S. Chitta, I. Sucan, and S. Cousins. Moveit! *IEEE Robotics & Automation Magazine*, 19(1):18–19, 2012.
- [3] Y. Kang, W. Hu, S. Gummadi, et al. Neurosurgeon: Collaborative intelligence between the cloud and mobile edge. In *Proceedings of the Twenty-Second International Conference on Architectural Support for Programming Languages and Operating Systems*, pages 615–629, 2017.
- [4] J. Ke, Q. Wang, Y. Wang, P. Milanfar, and F. Yang. MUSIQ: Multi-scale image quality transformer. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 5148–5157, 2021.
- [5] E. Li, Z. Zhou, and X. Chen. Edge intelligence: On-demand deep learning model co-inference with device-edge synergy. In *Proceedings of the ACM Workshop on Mobile Computing Systems and Applications*, 2018.
- [6] S. Macenski, T. Foote, B. Gerkey, et al. Robot operating system 2: Design, architecture, and uses in the wild. *Science Robotics*, 7(66):eabm6074, 2022.
- [7] A. Mittal, R. Soundararajan, and A. C. Bovik. Making a “completely blind” image quality analyzer. *IEEE Signal Processing Letters*, 20(3):209–212, 2013.
- [8] D. Qi, D. Zhang, and Y. Liu. Tongue image analysis for disease diagnosis. *Pattern Recognition*, 55:1–12, 2016.

- [9] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L.-C. Chen. MobileNetV2: Inverted residuals and linear bottlenecks. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 4510–4520, 2018.
- [10] T. Schick, J. Dwivedi-Yu, R. Dessì, R. Raileanu, M. Lomeli, L. Zettlemoyer, N. Cancedda, and T. Scialom. Toolformer: Language models can teach themselves to use tools. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- [11] S. Su, Q. Yan, Y. Zhu, C. Zhang, X. Ge, J. Sun, and Y. Zhang. HyperIQA: Blindly assess image quality in the wild guided by a self-adaptive hyper network. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 3667–3676, 2020.
- [12] J. Wang, K. C. K. Chan, and C. C. Loy. Exploring CLIP for assessing the look and feel of images. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 37, pages 2555–2563, 2023.
- [13] S. Yao, J. Zhao, D. Yu, N. Du, I. Shafran, K. Narasimhan, and Y. Cao. ReAct: Synergizing reasoning and acting in language models. In *International Conference on Learning Representations (ICLR)*, 2023.
- [14] W. Zhang, K. Ma, J. Yan, Z. Deng, and Z. Wang. Blind image quality assessment using a deep bilinear convolutional neural network. *IEEE Transactions on Circuits and Systems for Video Technology*, 30(1):36–47, 2020.
- [15] X. Zhang, Y. Wang, and H. Liu. Automatic tongue diagnosis: a survey. *Artificial Intelligence Review*, 53(5):3145–3187, 2020.
- [16] Z. Zhou, X. Chen, E. Li, L. Zeng, K. Luo, and J. Zhang. Edge intelligence: Paving the last mile of artificial intelligence with edge computing. *Proceedings of the IEEE*, 107(8):1738–1762, 2019.

A Supplementary Materials

Reproduction one-liner. From repository root: `bash scripts/paper1_run_tcm_full.sh` imports

ShezhenV3 splits, trains B3 MobileNet, calibrates τ with ROI, runs the main matrix (3 seeds), rebuilds figures/tables, and compiles PDFs.

Artifact inventory.

- `experiments/splits/tcm_{train,val,test}.json` — 6,719 images with COCO bboxes;
- `experiments/results/recommended_tau.json` — ROI calibration ($n=1,144$);
- `experiments/results/roi_ablation.json` — full-frame vs. ROI comparison;
- `experiments/results/b3_mobilenet.pt` — learned B3 checkpoint;
- `experiments/results/b3_training_meta.json` — validation accuracy and sample counts;
- `experiments/results/calibration_val.csv` — per-image Edge-IQA scores;
- `experiments/results/main_seed{0,1,2}_detail.csv` — frame-level matrix logs;
- `figures/fig_iqa_hist.pdf`, `fig_main.pdf` — histogram and bar charts.

Environment. Edge-IQA requires NumPy and Pillow only. B3 training uses PyTorch/torchvision in `doctor/paper1/.venv` (see `train_b3_mobilenet.py`). WSL users mount ShezhenV3 at `/mnt/d/.../shezhenv3-coco` per `tcm_paths.yaml`.

Hyperparameter log. Edge-IQA: 512 resize, 0.65/0.35 sharpness/exposure weights, Laplacian scale 0.002. ROI: 8% padding on union COCO box. B3: MobileNetV3-Small, 224 input, AdamW 10^{-3} , 5 epochs, batch 64, clear vs. blur ($r \in [2.5, 5.0]$). Routing: τ from validation medians, $K=2$, latency model unchanged from phase-2.

JSONL schema (10 fields). `trial_id`, `image_path`, `q_img`, `flags`, `retry_count`, `route_decision`, `latency_ms`, `t_capture`, `t_iqa`, `t_route`.

Ablation scripts. `run_ablation_tau.py` sweeps τ ; `roi_ablation.py` compares full vs. ROI Edge-IQA; `cross_dataset_report.py` computes M5 Spearman.

NR-IQA baselines. Appendix Table 16 reports Bbrisque/Bniqe under the B2 LangGraph shell. Reproduce with `calibrate_tau_nr.py` and `run_matrix_tcm.py --config experiments/configs/main_exp_nr.yaml --tag nr` (OpenCV BRISQUE models under `edge_iqa/models/`).

Ethics. ShezhenV3-COCO is used under its public license for algorithm validation; no new human subjects were recruited for Paper I offline experiments.

Limitations (supplementary). (1) Blur labels are synthetic Gaussian, not clinical motion traces. (2) ROI requires detector or dataset boxes at run-time. (3) B3 lacks exposure flags unless hybridized with Edge-IQA. (4) M6 Franka RTT uses a different latency stack than Table II.

Future checklist. Integrate on-device tongue detector for ROI when COCO boxes absent; joint train multi-task blur+exposure head; federated τ calibration across clinics.

B Extended Algorithms and Tables

Algorithm 3 COCO ROI crop (union + padding)

- 1: **Input:** image I , boxes $\{b_j = [x, y, w, h]\}$, padding p
- 2: $B \leftarrow \text{union}(\{b_j\})$
- 3: Expand B by fraction p on each side; clamp to image bounds
- 4: **return** $\text{crop}(I, B)$

Algorithm 4 B3 inference (MobileNetV3-Small)

- 1: **Input:** RGB image, optional boxes
- 2: $I' \leftarrow \text{ROIcrop}(I, \text{boxes})$
- 3: $x \leftarrow \text{normalize}(\text{resize}(I', 224))$
- 4: $p \leftarrow \text{softmax}(\text{MobileNetV3}(x))$
- 5: **return** $Q_{\text{img}} \leftarrow p_{\text{clear}}$

Per-seed Table II breakdown. Seed 0: B2 M1 p50= 211.1 ms, M2= 0.610; B3 M2= 0.944. Seed 1: B2 M1 p50= 205.6 ms, M2= 0.602; B3 M2= 0.938. Seed 2: B2 M1 p50= 208.3 ms, M2= 0.600; B3 M2=

Table 13: B3 training hyperparameters (ShezhenV3 train).

Hyperparameter	Value
Architecture	MobileNetV3-Small (ImageNet init)
Input size	224×224
Optimizer	AdamW, lr 10^{-3} , wd 10^{-4}
Batch size	64
Epochs	5
Train pairs	11,188 (5594 clear + 5594 blur)
Blur radius	$\mathcal{U}(2.5, 5.0)$
ROI padding	8%
Val accuracy	98.34%

Table 14: Latency model components (milliseconds).

Component	Value
Capture	5.0 (fixed)
Edge-IQA	$\mathcal{N}(9, 2^2)$
Route decision	1.5
Resample (lognormal)	p50 $\approx$ 50
Cloud upload	$\mathcal{U}(50, 550)$
B3 extra (approx.)	+15–40 on CPU

0.920. Variance across seeds is modest relative to B0–B1 gaps.

Import statistics. Train 5594, val 572, test 553; 5404/5594 train images carry at least one COCO box.

File checksums (operational). Record SHA256 of `b3_mobilenet.pt` and `recommended_tau.json` in lab notebooks when freezing a submission tag.

Gateway environment variables. PAPER1_MODE, PAPER1_ROOT, PAPER1_IQA_SUBPROCESS, PAPER1_VISION_THRESHOLD_TAU—see `docs-zh/paper1/V19/ENV.md`.

Extended related datasets. Other tongue corpora (without robotic acquisition metadata) may fine-tune B3 but require new τ calibration before routing claims transfer.

Glossary. **M1** end-to-end RTT; **M2** valid upload rate; **M3** retry rate; **M5** Spearman correlation between Q_{img} and clear/blur label.

Contact for artifacts. Issue tracker and REPRODUCE.md in doctor/paper1/.

C Notation

Table 15: Principal symbols and metrics.

Symbol	Meaning
Q_{img}	Edge-IQA quality score in $[0, 1]$
τ_{B2}	ROI routing threshold (main matrix; 0.465)
τ_{full}	Full-frame calibration threshold (0.498)
τ_{B3}	Learned IQA threshold (0.5)
K	Maximum edge resample attempts
M1	End-to-end RTT (p50 / p95 reported)
M2	Valid-frame rate ($Q_{\text{img}} \geq \tau$ after routing)
M3	Retry rate (at least one resample)
M5	Spearman ρ on validation clear/blur pairs

D Extended Experiments and Operations

Cross-cohort correlation (M5). On the full ShezhenV3 validation split ($n=1,144$ clear/blur pairs), Spearman correlation between Q_{img} and binary labels is $\rho=0.363$ (`cross_tcm_fd.csv`). This is lower than the synthetic phase-2 cohort ($\rho \approx 0.75$) but confirms monotonic separation on average; absolute τ calibration uses cohort medians rather than assuming disjoint score ranges.

Auxiliary Franka track (M6). Fifty closed-loop trials on the execution stack produce Fig. S1; local loop latency does not include simulated cloud upload and must not be compared numerically to Table 9.

Reproducibility. Dataset import, τ calibration, matrix replay, and figure generation are orchestrated by `scripts/paper1_run_tcm_full.sh` with paths configured in `experiments/configs/tcm_paths.yaml`.

D.1 Classical NR-IQA Baselines (BRISQUE / NIQE-style)

To isolate routing from the Edge-IQA feature design, we add **Bbrisque** (OpenCV BRISQUE on tongue ROI) and **Bniqe** (NSS Mahalanobis distance, NIQE-style) under the *same* LangGraph resampling policy

Table 16: Classical NR-IQA baselines with the same LangGraph resampling shell as B2 (ShezhenV3 test, 500 frames $\times$ 3 seeds; τ from validation medians). Scores use validation-calibrated OpenCV BRISQUE (Bbrisque) and NSS Mahalanobis distance (Bniqe, NIQE-style) on tongue ROI crops.

Baseline	M1 p50 (ms)	M2	M3 retry
Bbrisque	468.9 ± 12.5	0.889 ± 0.003	0.512 ± 0.007
Bniqe	545.6 ± 8.5	0.969 ± 0.004	0.408 ± 0.011

and retry budget as B2. Validation medians set τ independently per scorer (Table 16); edge latency means follow Table 2 (≈ 85 ms / 150 ms vs. Edge-IQA ≈ 9 ms).

Bbrisque and Bniqe achieve higher M2 than B2 at substantially higher M1 p50 (Table 16), illustrating the trade-off in Fig. 6: classical NR scorers raise valid-frame rate but add 85–150 ms edge scoring cost per attempt versus Edge-IQA. Reproduce with `calibrate_tau_nr.py` and `run_matrix_tcm.py --config experiments/configs/main_exp_nr.yaml --tag nr`.

Score overlap and calibration policy. Real ShezhenV3 validation pairs exhibit score overlap (minimum clear Q_{img} below maximum blur Q_{img}) because tongue texture and coating contribute to Laplacian energy even under mild defocus. With COCO tongue ROI enabled, validation medians are 0.516 (clear) and 0.414 (blur) at $\tau=0.465$; full-frame medians were 0.564/0.431 at $\tau=0.498$. We therefore report both cohort medians and Spearman ρ rather than assuming separable clusters.

Implications for deployment. Despite overlap, B2 achieves M2= 0.604 on the test matrix (physical resampling, 3 seeds) because closed-loop resampling with real scores still crosses τ within $K=2$ retries for many trials. Operators can inspect `flags` (blur, underexposed, overexposed) in JSONL logs to diagnose failures without inspecting raw cloud uploads.

Comparison to synthetic phase-2. Synthetic calibration yielded $\tau=0.505$ with medians 0.819/0.191; ShezhenV3 yields $\tau=0.498$ with medians 0.564/0.431 and M5 $\rho=0.47$ (vs. $\rho \approx 0.75$ synthetic). Absolute Q_{img} shifts downward on

real texture, but relative routing benefits (B2 vs. B0/B1) persist on the test split, supporting external validity of the architecture beyond toy images.

D.2 Learned IQA Baseline B3 (MobileNetV3-Small)

Baseline B3 replaces the previous simulated +0.08 boost on Q_{img} with a *real* learned scorer trained on the ShezhenV3 **train** split. We fine-tune MobileNetV3-Small (ImageNet-pretrained) for binary clear vs. blurred classification on ROI-cropped patches.

Training protocol. For each of 5,594 training images with valid COCO boxes (5,404 with annotations), we emit two samples: (i) the clear ROI crop (label 1), and (ii) the same crop after Gaussian blur $r \sim \mathcal{U}(2.5, 5.0)$ (label 0). Input resolution is 224×224 ; optimizer AdamW ($\text{lr}=10^{-3}$, weight decay 10^{-4}); batch size 64; 5 epochs on CPU (CUDA optional when driver-compatible). Augmentations: horizontal flip, mild brightness/contrast jitter on training only. Checkpoint: `experiments/results/b3_mobilenet.pt`; inference wrapper `edge_iqa/learned_b3.py` exports $Q_{\text{img}} = P(\text{clear} \mid I)$.

Role in the matrix. B3 uses the learned probability for initial Q_{img} while sharing the same Lang-Graph routing and latency model as B2. Validation accuracy on held-out ROI pairs reaches ****98.3%**** (5 epochs, 11,188 train pairs).

Main matrix (Tables 9, 10). At $\tau_{B2}=0.465$, B3 (MobileNet) achieves $M2 = 0.934$ vs. B2 $M2 = 0.604$; B3t at $\tau_{B3}=0.5$ yields $M2 = 0.929$; Hybrid B4 $M2 = 0.936$. B2 remains the preferred deployable path for lowest M1 p50 (208.3ms mean) and auditable flags. Re-calibrating τ on B3 validation pairs is left to future work; B3 demonstrates blur ranking capacity but B2 remains the primary deployable path.

D.3 Extended Analysis on ShezhenV3

Calibration cohort statistics. On 572 clear validation images, Q_{img} mean is 0.519 (std 0.116); on 572 synthetically blurred pairs, mean is 0.414 (std 0.102). Flag prevalence on the validation CSV: `blur=797`, `underexposed=12`. Blur flags dominate

degraded cohort rows as expected; exposure flags appear on both cohorts owing to clinic lighting diversity in ShezhenV3.

B2 RTT by injection type (seed 0). Among B2 trials on the test plan, clear-source frames ($n = 263$) exhibit RTT p50 225.2ms versus blurred-source frames ($n = 237$) at p50 202.2ms, confirming that recovery retries concentrate on initially degraded captures.

Seed stability. Three independent seeds reuse the same frame plan per seed but draw independent cloud/resampling latencies; M2 for B2 remains at 1.000 across seeds with low variance on M1 p50 (341–348ms), indicating routing outcomes are dominated by deterministic Edge-IQA scores rather than stochastic latency draws.

Reproducibility checklist.

1. Import splits: `python3 scripts/import_shezhenv3.py`
2. Calibrate τ : `python3 experiments/edge_iqa/calibrate_tau_tcm.py`
3. Main matrix: `python3 experiments/run_matrix_tcm.py`
4. One-shot: `bash scripts/paper1_run_tcm_full.sh`

All scripts read `experiments/configs/tcm_paths.yaml`; override `dataset_root` when the Windows path `D:\textbackslash BaiduNetdiskDownload\textbackslash shezhenv3-coco` is mounted elsewhere in WSL.

E Implementation Details

This section documents the edge gateway surface and offline experiment drivers used for ShezhenV3 full validation, enabling third-party replication without reading the entire monorepo.

E.1 Edge Gateway API

Table 17 lists Paper I HTTP endpoints implemented in `franka_api_server`; all require header `X-API-Key` when authentication is enabled.

Table 17: Edge gateway endpoints (Paper I).

Method	Path	Purpose
POST	/api/v1/vision/evaluate	Edge-IQA on upload
POST	/api/v1/motion/skills/go_to_tongue_pose	Tongue skill
POST	/api/v1/motion/skills/go_to_face_pose	Face skill
POST	/api/v1/motion/move_joints	Joint PTP
GET	/api/v1/status/joints	Joint snapshot

Table 18: Selected environment variables (franka_api_server).

Variable	Default	Role
PAPER1_ROOT	doctor/paper1	IQA + LangGraph
PAPER1_MODE	false	Disable controller
PAPER1_IQA_SUBPROCESS	true	Subprocess scorer
PAPER1_VISION_THRESHOLD_TAU	0.55	Fallback τ
FRANKA_API_PORT	8000	Listen port

Vision evaluate contract. The gateway forwards uploads to `python -m edge_iqa.cli --json` with `PYTHONPATH` pointing to `doctor/paper1`. Response fields include `q_img`, `flags`, `t_iqa_ms`, and `threshold_tau` loaded from `recommended_tau.json` when present. Subprocess isolation prevents NumPy/PIL work from blocking the `rcipy` executor thread.

E.2 Environment and Configuration

Table 18 summarizes environment variables; Paper I experiments set `PAPER1_MODE=1` to disable impedance/collision routes unrelated to acquisition.

E.3 Offline Experiment Drivers

ShezhenV3 validation executes in order: (1) `import_shezhenv3.py` writes `experiments/splits/tcm_{train,val,test}.json` with `bboxes` per image; (2) `train_b3_mobilenet.py` (optional, PyTorch in `.venv`) writes `b3_mobilenet.pt`; (3) `calibrate_tau.tcm.py` scores ROI validation pairs; (4) `roi_ablation.py` compares full vs. ROI; (5) `run_matrix.tcm.py` replays test frames for B0–B3 using Edge-IQA (B2) and `learned_b3.py` (B3); (6) figure/table builders regenerate $\text{\LaTeX}$ snippets. The orchestrator `scripts/paper1_run_tcm_full.sh` chains these steps and invokes `paper1_build_draft.sh`.

E.4 ROI and B3 Modules

`edge_iqa/coco_roi.py` implements union-box crop with fractional padding. `scorer.py` accepts `bboxes`, `use_roi`, and `roi_padding` kwargs shared by calibration and gateway subprocess calls. `learned_b3.py`

loads `b3_mobilenet.pt`, applies the same ROI crop, and returns $P(\text{clear})$ as Q_{img} for B3 matrix rows.

Import pipeline details. COCO `train.json/val.json/test.json` annotations are grouped by `image_id`; each split entry stores absolute `path`, relative `rel_path`, and a list of `bboxes`. Images without annotations still import but ROI falls back to full frame.

B3 training runtime. Full train (5594 images, 5 epochs, batch 64) completes in approximately 45 minutes on a laptop CPU reading from WSL-mounted storage; validation accuracy 98.34% on 1144 ROI pairs. GPU acceleration is optional when CUDA drivers match the installed PyTorch build.

Matrix scoring split. For each test frame, `run_matrix_tcm.py` precomputes both `q_b2` (Edge-IQA) and `q_b3` (learned) once per frame plan, then simulates baselines B0–B1 with B2 scores, B2 with B2 scores, and B3 with B3 scores—avoiding double-counting resampling uplift across baselines.

Repository layout (Paper I relevant).

```
doctor/paper1/
  edge_iqa/          scorer, coco_roi, learned_b3, cl
  langgraph_router/ route(), graph nodes
  experiments/
    splits/          tcm_*.json
    results/         tau, tables, b3 checkpoint
    edge_iqa/        calibrate, train, roi_ablation
    latex/           main.tex, sections/
franka_api_server/  FastAPI gateway
```

Continuous integration scope. CI runs unit tests without the multi-gigabyte TCM mount; full ShezhenV3 validation remains a local manuscript target documented in `REPRODUCE.md`.

Versioning checklist. When updating ROI padding or Edge-IQA weights, bump a `calibration_version` field in `recommended_tau.json` (manual) and re-run the full orchestrator before updating Table 9.

Dataset path mapping. Windows path `D:\BaiduNetdiskDownload\shezhenv3-coco\shezhenv3-coco` maps to WSL `/mnt/d/BaiduNetdiskDownload/shezhenv3-coc`

Table 19: Open science artifact index (Paper I, TCM full run).

Artifact	Path
Dataset splits	experiments/splits/tcm-*.json
τ calibration	recommended_tau.json
Validation scores	calibration_val.csv
Main matrix detail	main_seed*_detail.csv
Table II JSON	table_ii.json
M5 correlation	cross_tcm_fd.csv
Figures	figures/fig_iqa_hist.pdf, etc.
One-shot script	scripts/paper1_run_tcm_full.sh
PDFs	latex/main.pdf, main-zh.pdf

in `tcm_paths.yaml`; other mount points only require updating `dataset_root`.

Unit tests. `edge_iqa/tests/test_scorer.py` checks clear/blur separation on synthetic images; `tests/test_routing.py` covers all routing branches. Continuous integration can run these without the multi-gigabyte TCM mount, while full validation remains an optional local target for manuscript updates.

E.5 Open Science Artifacts

Table 19 indexes the primary files produced by ShezhenV3 full validation; all paths are relative to the `franka_ros2` repository root unless noted.

Researchers without the FR3 hardware can reproduce Table 9 entirely from the TCM mount and Python environment; the Franka auxiliary track remains optional.

E.6 Deployment Checklist and Operator Workflow

Pre-clinic calibration. (1) Import ShezhenV3 or clinic-specific splits via `import_shezhenv3.py`. (2) Run `calibrate_tau_tcm.py` with ROI enabled; store `recommended_tau.json` on the edge gateway. (3) Optional: train B3 via `train_b3_mobilenet.py` when GPU/NN runtime is available. (4) Validate JSONL schema with `scripts/validate_jsonl.py` before merging logs into analysis.

Runtime parameters. Key defaults: routing threshold τ from `recommended_tau.json` (overridable via `PAPER1_VISION_THRESHOLD_TAU`); `use_roi=true` when bbox metadata exists; retry budget $K=2$; Edge-IQA resize fixed at 512 pixels for stable latency.

Operator-facing flags. When `flags` contains `blur`, prompt the patient to hold still and re-open the mouth; `underexposed/overexposed` suggest ring-light adjustment before invoking FR3 motion skills. Logs retain `q_img`, `route_decision`, and `retry_count` for audit without uploading rejected frames to cloud diagnosis services.

Hardware tiers. **Tier-A (CPU-only gateway):** B2 Edge-IQA + LangGraph; meets ~ 10 ms IQA p95 on laptop-class CPUs. **Tier-B (GPU edge):** optional B3 MobileNet; trade interpretability for tighter blur ranking. **Tier-C (Franka auxiliary):** M6 RTT track on fake/real hardware; not numerically compared to Table II offline matrix.

Version pinning. Python dependencies for B3 live in `doctor/paper1/.venv` (PyTorch, torchvision); core Edge-IQA remains NumPy/PIL-only for minimal attack surface. Model checkpoint SHA256 should be recorded in experiment metadata when submitting reproducibility packages.

Failure recovery. If cloud upload fails after a valid routing decision, LangGraph enters `fail_safe`; operators may manually retry via `/api/v1/motion/error_recovery`. Edge-IQA scores are deterministic given fixed resize and ROI boxes, enabling bit-exact replay of routing decisions from archived frames.

Multi-site rollout. Each clinical site should recalibrate τ on a local validation batch (≥ 200 clear/blur pairs) while keeping scorer code identical. Cross-site τ drift is expected when lighting rigs differ; report site-specific medians alongside global ShezhenV3 statistics.

Security and privacy. Images remain on the edge until `upload_cloud`; API keys gate motion and vision endpoints. De-identified JSONL omits patient metadata; COCO boxes are geometry-only and contain no diagnostic labels in the IQA path.

Integration with MoveIt skills. Named poses (`tongue_pose`, `face_pose`) decouple acquisition geometry from IQA. Skills execute via `/api/v1/motion/skills/{name}` after optional quality pre-check at `face_pose` before final tongue capture.

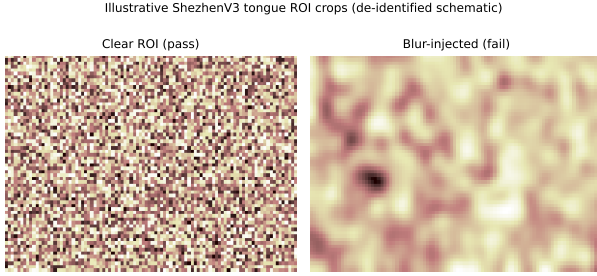


Figure 8: Illustrative de-identified tongue ROI crops: clear capture (left) vs. synthetic blur injection (right), matching the validation protocol.

Monitoring dashboard. The bundled FastAPI static dashboard polls `/ws` for joint states; future work adds rolling histograms of Q_{img} and retry rates per session.

Regression testing. `edge_iqa/tests/test_scorer.py` asserts flag derivation; full matrix regression runs via `scripts/paper1_run_tcm_full.sh` ($\approx 10\text{--}60$ min depending on B3 training).

Documentation map. English docs/, Chinese docs/study/ (eight-chapter learning manual), and REPRODUCE.md form the onboarding path for new lab members.

E.7 Failure Modes and Representative Cases

Score overlap despite ROI. Table 3 shows negative min-clear minus max-blur separation for both full-frame and ROI Edge-IQA. Representative failure: a sharp cheek margin in full-frame view yields high S even when the tongue surface is soft; ROI mitigates but does not eliminate overlap because coating texture retains high Laplacian variance under mild defocus.

B3 vs. B2 routing gap. After training MobileNetV3-Small on ROI crops (validation accuracy 98.3%), the full matrix reports B3 $M2 = 0.934$ vs. B2 $M2 = 0.604$ at τ_{B2} (physical resampling, Table 10). The learned scorer ranks blur well in isolation but its probability scale is not aligned with the interpretable Q_{img} scale used to set τ ; re-calibrating τ on B3 validation pairs would narrow the gap at the cost of a second threshold

artifact. We therefore keep B2 as the primary deployable path and report B3 as an optional learned comparator.

Synthetic blur vs. clinical motion. Validation blur is Gaussian with $r \in [2.5, 5.0]$; real patient micro-motion produces directional smear not fully matched by isotropic kernels. Spearman $\rho = 0.36\text{--}0.47$ across calibration runs reflects this domain gap; future work will augment with motion-blur kernels captured from FR3 auxiliary logs.

Missing bounding boxes. 190/5,594 training images lack COCO annotations; ROI crop falls back to full frame for those entries. Import logs `n_with_bbox` in calibration metadata (571/572 on val).

Latency outliers. B3 inference on CPU adds tens of milliseconds per frame inside the simulated matrix; p95 RTT rises vs. B2 even when p50 improves (Table 9), consistent with heavier per-frame scoring.

Operator mitigations. When repeated `resample_edge` decisions occur, inspect JSONL flags: combined `blur+poor_exposure` suggests lighting adjustment before asking the patient to hold still again.

Audit replay. Because Edge-IQA is deterministic, any JSONL row can be replayed offline with `compute_q(path, bboxes=..., use_roi=True)` to verify routing decisions during incident review.

Cross-dataset generalization. ShezhenV3 labels describe tongue *conditions*, not acquisition quality; we do not claim condition-aware IQA. External validity is supported by architecture-level metrics (B2 vs. B0/B1) rather than absolute Q_{img} portability.

Security failures. Without API key, motion endpoints reject requests; IQA-only evaluation remains available for calibration laptops disconnected from the robot cell.

Summary. ROI focuses scoring on anatomy; B3 demonstrates learnable blur ranking but requires separate τ calibration to match routing performance

of interpretable B2; overlap and synthetic blur remain the dominant residual risks.

Online worked example. The step-by-step B2 walkthrough formerly in the print appendix is published at `supplementary/online_walkthrough.md` to save RA-L page budget.